

100 Linux Commands for Freshers

Here are 100 Linux commands for freshers, grouped by practical use.

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1. Basic Navigation

Command	Description
<code>pwd</code>	Show current directory.
<code>ls</code>	List files and folders.
<code>ls -l</code>	List with details.
<code>ls -a</code>	Show hidden files.
<code>ls -lh</code>	Show file size in human-readable format.
<code>cd /path</code>	Change directory.
<code>cd ..</code>	Go one directory back.
<code>cd ~</code>	Go to home directory.
<code>clear</code>	Clear terminal screen.
<code>history</code>	Show command history.

File and Directory Management

Command	Description
<code>touch file.txt</code>	Create an empty file.
<code>mkdir mydir</code>	Create a directory.
<code>mkdir -p dir1/dir2</code>	Create nested directories.
<code>cp file1 file2</code>	Copy file.
<code>cp -r dir1 dir2</code>	Copy directory.
<code>mv file1 file2</code>	Rename or move file.
<code>rm file.txt</code>	Remove file.
<code>rm -r dir</code>	Remove directory.
<code>rm -rf dir</code>	Force remove directory.
<code>rmdir emptydir</code>	Remove empty directory.

Viewing File Content

Command	Description
<code>cat file.txt</code>	Show full file content.
<code>less file.txt</code>	View file page by page.
<code>more file.txt</code>	View file page by page.
<code>head file.txt</code>	Show first 10 lines.
<code>head -n 20 file.txt</code>	Show first 20 lines.
<code>tail file.txt</code>	Show last 10 lines.
<code>tail -n 50 file.txt</code>	Show last 50 lines.
<code>tail -f file.log</code>	Watch live log updates.
<code>wc -l file.txt</code>	Count lines.
<code>nl file.txt</code>	Show file with line numbers.

File Search

Command	Description
<code>find / -name file.txt</code>	Find file by name.
<code>find . -type f</code>	Find files in current directory.
<code>find . -type d</code>	Find directories.
<code>find . -name "*.log"</code>	Find all .log files.
<code>locate file.txt</code>	Quick file search.
<code>which ls</code>	Show command path.
<code>whereis bash</code>	Find binary/source/manual location.
<code>grep "error" file.log</code>	Search text in file.
<code>grep -i "error" file.log</code>	Case-insensitive search.
<code>grep -r "password" /etc</code>	Recursive search.

Text Processing

Command	Description
<code>sort file.txt</code>	Sort lines.
<code>uniq file.txt</code>	Remove duplicate adjacent lines.
<code>sort file.txt uniq</code>	Sort and remove duplicates.
<code>cut -d':' -f1 /etc/passwd</code>	Extract first field.
<code>awk '{print \$1}' file.txt</code>	Print first column.
<code>sed 's/old/new/g' file.txt</code>	Replace text.
<code>tr 'a-z' 'A-Z'</code>	Convert lowercase to uppercase.
<code>diff file1 file2</code>	Compare two files.
<code>cmp file1 file2</code>	Compare files byte by byte.
<code>tee file.txt</code>	Save command output to file.

Permissions and Ownership

Command	Description
<code>chmod 755 file.sh</code>	Change file permission.
<code>chmod +x script.sh</code>	Make file executable.
<code>chmod -R 755 dir</code>	Change permission recursively.
<code>chown user file.txt</code>	Change file owner.
<code>chown user:group file.txt</code>	Change owner and group.
<code>chown -R user:group dir</code>	Change ownership recursively.
<code>ls -l</code>	Check permissions.
<code>umask</code>	Show default permission mask.
<code>stat file.txt</code>	Show file details.
<code>id</code>	Show current user UID/GID.

User and Group Management

Command	Description
<code>whoami</code>	Show current username.
<code>who</code>	Show logged-in users.
<code>w</code>	Show logged-in users and activity.
<code>id username</code>	Show user details.
<code>useradd username</code>	Create user.
<code>passwd username</code>	Set user password.
<code>usermod -aG group username</code>	Add user to group.
<code>groupadd groupname</code>	Create group.
<code>su - username</code>	Switch user.
<code>sudo command</code>	Run command as root.

Process Management

Command	Description
ps	Show current shell processes.
ps aux	Show all running processes.
top	Live process monitor.
htop	Interactive process monitor.
pidof nginx	Find process ID.
pgrep nginx	Search process by name.
kill PID	Kill process by PID.
kill -9 PID	Force kill process.
pkill nginx	Kill process by name.
jobs	Show background jobs.

Disk and Storage

Command	Description
<code>df -h</code>	Show disk usage.
<code>du -sh *</code>	Show folder/file sizes.
<code>du -sh /var/log</code>	Show directory size.
<code>lsblk</code>	List block devices.
<code>blkid</code>	Show disk UUIDs.
<code>mount</code>	Show mounted filesystems.
<code>mount /dev/sdb1 /mnt</code>	Mount disk.
<code>umount /mnt</code>	Unmount disk.
<code>fdisk -l</code>	Show disk partitions.
<code>free -h</code>	Show memory usage.

Networking

Command	Description
<code>ip addr</code>	Show IP address.
<code>ip route</code>	Show routing table.
<code>ping google.com</code>	Check connectivity.
<code>curl https://example.com</code>	Fetch URL content.
<code>wget https://example.com/file.zip</code>	Download file.
<code>ss -tulnp</code>	Show listening ports.
<code>netstat -tulnp</code>	Show listening ports.
<code>hostname</code>	Show hostname.
<code>hostname -I</code>	Show system IP address.
<code>nslookup google.com</code>	DNS lookup.

Package Management

RHEL / AlmaLinux / CentOS

Command	Description
<code>dnf install package</code>	Install package.
<code>dnf remove package</code>	Remove package.
<code>dnf update</code>	Update packages.
<code>rpm -qa</code>	List installed RPM packages.
<code>rpm -qi package</code>	Show package information.

Ubuntu / Debian

Command	Description
<code>apt update</code>	Refresh package list.
<code>apt install package</code>	Install package.
<code>apt remove package</code>	Remove package.
<code>apt upgrade</code>	Upgrade packages.
<code>dpkg -l</code>	List installed packages.

Services and Logs

Command	Description
<code>systemctl status nginx</code>	Check service status.
<code>systemctl start nginx</code>	Start service.
<code>systemctl stop nginx</code>	Stop service.
<code>systemctl restart nginx</code>	Restart service.
<code>systemctl enable nginx</code>	Enable service at boot.
<code>systemctl disable nginx</code>	Disable service at boot.
<code>journalctl</code>	View system logs.
<code>journalctl -u nginx</code>	View logs for one service.
<code>journalctl -xe</code>	View recent error logs.
<code>dmesg</code>	Show kernel messages.

Compression and Archive

Command	Description
<code>tar -cvf archive.tar dir</code>	Create tar archive.
<code>tar -xvf archive.tar</code>	Extract tar archive.
<code>tar -czvf archive.tar.gz dir</code>	Create gzip tar archive.
<code>tar -xzvf archive.tar.gz</code>	Extract gzip tar archive.
<code>zip file.zip file.txt</code>	Create zip file.
<code>unzip file.zip</code>	Extract zip file.

Useful Admin Commands

Command	Description
<code>date</code>	Show date and time.
<code>uptime</code>	Show system uptime.
<code>reboot</code>	Restart system.
<code>shutdown now</code>	Shutdown system.
<code>shutdown -r now</code>	Restart system.
<code>uname -a</code>	Show kernel and OS info.
<code>cat /etc/os-release</code>	Show Linux OS version.
<code>env</code>	Show environment variables.
<code>export VAR=value</code>	Set environment variable.
<code>crontab -e</code>	Edit cron jobs.

Bonus: Most Important Commands for Freshers

Learn these first:

`pwd ls cd mkdir touch cp mv rm cat less head tail`
`grep find chmod chown ps top df -h du -sh free -h ip`
`addr ping curl systemctl journalctl tar sudo`

These commands are enough to start working confidently on Linux servers.

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